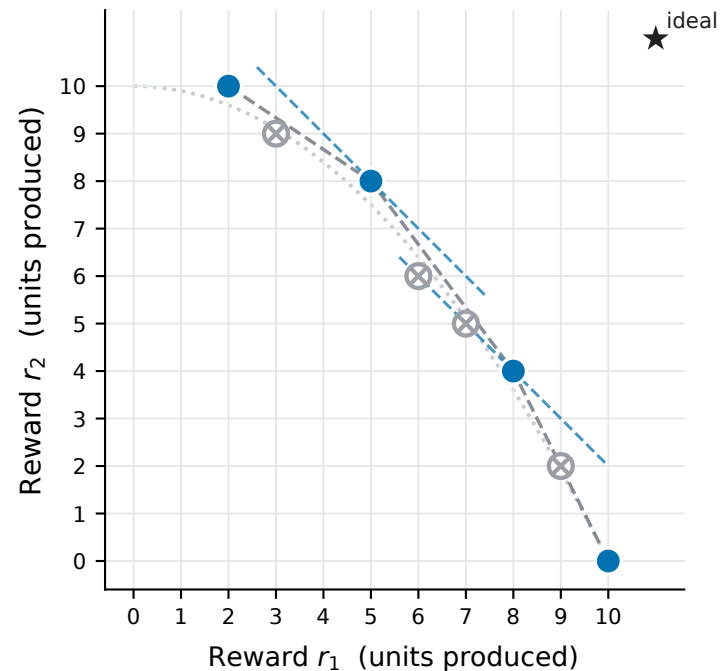


Discrete Reward Landscapes: Linear Scalarization Misses Integer Pareto Points that Reference-Point Scalarization Recovers

Linear scalarization

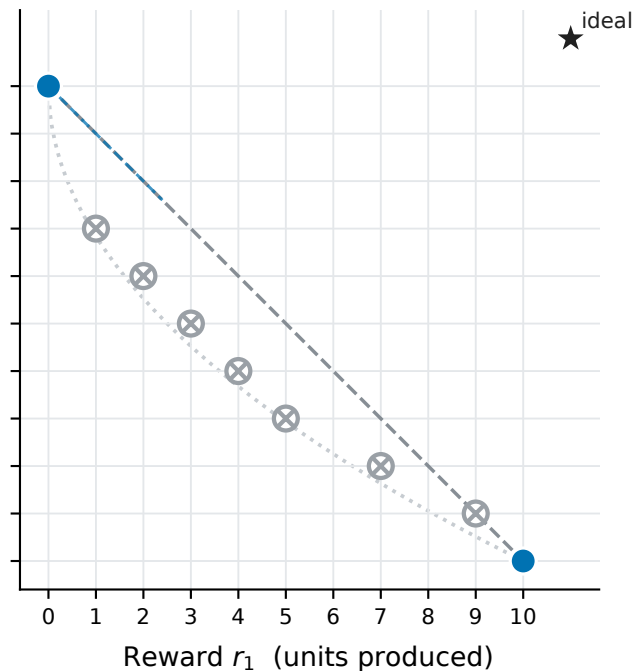
Convex

reaches 4 / 8 Pareto points



Non-convex

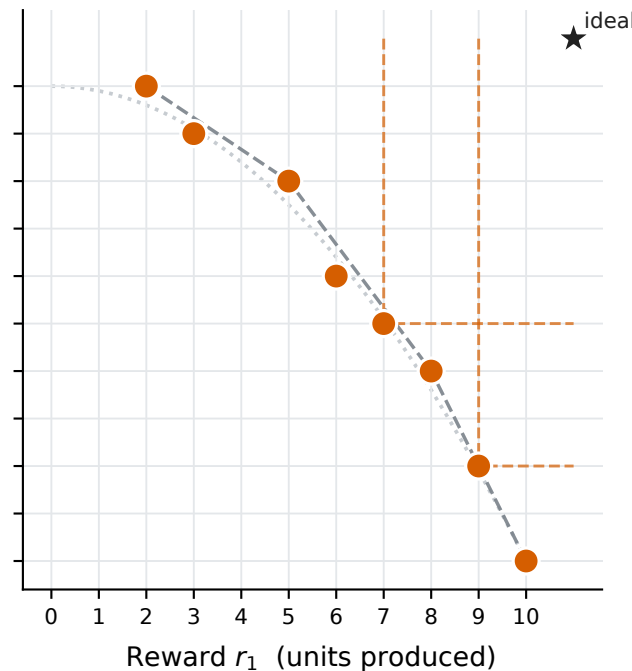
reaches 2 / 9 Pareto points



Smooth Tchebycheff

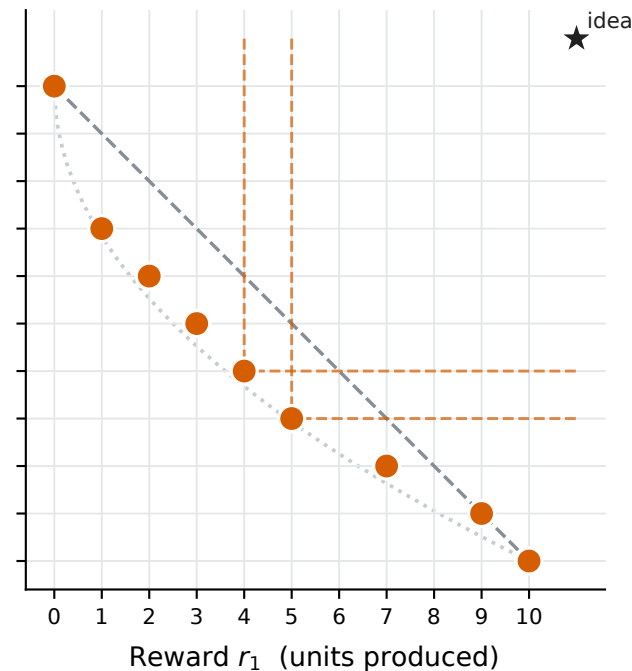
Convex

reaches 8 / 8 Pareto points



Non-convex

reaches 9 / 9 Pareto points



● Reachable optimum × Linearly unreachable Pareto point --- Convex hull of attainable set Continuous relaxation (reference) ★ Reference / ideal point